Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years of experience building **React** products, **REST APIs**, and **SQL** integrations; improving **reliability** and latency across production data workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing** and query optimization, designing normalized Azure SQL schemas and row-level isolation for a multi-tenant education platform serving 500+ users.
- Achieved 99%+ production reliability across 5K+ monthly integration jobs through **FastAPI data pipelines**, incremental loads, reconciliation checks, and CI/CD releases, connecting relational databases and cloud storage with validated transformations.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% for 150+ instructors through a **React and Flask** grading platform, connecting upload, rubric generation, evaluation, and history workflows while handling 10K+ daily grading requests.
- Enabled fault-tolerant processing of 1K+ files per batch using **Python multithreading**, secure PDF/DOCX/ZIP ingestion, REST APIs, and AWS RDS, supporting concurrent assessment workflows without blocking the instructor-facing user interface.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **pagination, filtering, and caching**, optimizing indexed queries and low-bandwidth payloads while integrating cross-platform services with object-oriented modules and maintainable API design patterns.
- Increased data pipeline efficiency by 68% through **AWS Glue and S3** transformations; automated executable builds and 20+ unit/integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Reduced resource wastage by 42% through a **React and Node.js** resource-management platform, defining microservices for allocation, customers, billing, and notifications while supporting WebSocket synchronization of 2M+ records across locations.
- Increased lead conversions by 15% through **CRM workflows** and usage analytics, integrating Azure SQL-backed records, role-based access, and real-time updates with Jenkins testing and Docker packaging for staged deployments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms Flask APIs** across 50K+ house-help booking records, implementing pricing, coupon validation, payment webhooks, and Firebase status synchronization while using transactions and locking to prevent conflicting time-slot reservations.
- Reduced manual refund handling by approximately 70% through **payment webhook automation**, supporting refunds, disputes, and consistent payment status while presenting end-to-end architecture and backend integration workflows directly to clients.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL
Frontend & APIs: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks
Data & Integration: PostgreSQL, Redis, MySQL, Azure SQL, MongoDB, Firebase, Data Modeling, ETL, Pagination, Caching
Cloud & Delivery: AWS EC2, AWS S3, AWS RDS, AWS Glue, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Engineering: Unit Testing, Integration Testing, PyTest, Mockito, Git, System Design, Data Validation, JWT, Fault Isolation

Projects

RecordBridge - Design Concept

- Proposed a **TypeScript connector** design to unify business records using shared schemas, pagination, authentication, and WebSockets, with React review screens and planned contract tests for cross-system sync recovery scenarios.

FinNexus | Live

- Enabled secure bank linking and **real-time financial workflows** across 1M+ transactions using Plaid, Next.js, TypeScript, and MongoDB, combining transaction filtering, token-based access controls, and Kubernetes-backed deployment for 50K+ users.

Chat2SQL | Code

- Unified **10+ database systems** in a Java and TypeScript workspace extending Chat2DB, integrating schema exploration, SQL generation, structure synchronization, and Docker-based local deployment across Windows, macOS, and Linux environments.